UNIVERSITY OF CALGARY

Impacts of STRs and STR Policy Tools on the Housing Market

Examination of the Causal Empirical Evidence

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Executive Summary

We examine the empirical literature that measures the impact of STR activity on different aspects of the housing market as well as how various policy tools impact STRs and different aspects of the housing market. There are several take-aways from a review of the empirical evidence:

- An increase in STR activity have been shown to have a positive causal impact on LTR prices and housing prices. The magnitude of this effect differs, but it generally suggests that increases in STR activity explains about from five to 20 percent of LTR rent increases and 14 percent to 33 percent of housing price increases.
- In areas with more STRs, units that shift to STRs are more likely to be "affordable housing units" (e.g., smaller units with less amenities) and units that would otherwise have sat vacant (e.g., whose hosts are away part-year so the unit could not be used as an LTR). Hosts of these units (e.g., otherwise vacant, small units) are more likely to be economically disadvantaged.
- Restrictions on STRs can significantly reduce STR listings. There is no evidence what proportion of those STR listings are put on the LTR market, sit vacant, or become owner-occupied.
- Restrictions on STRs affect LTR and house prices differently: there are distributional effects.
- Restricting STRs (e.g., outright prohibitions and/or personal residence requirements) can positively impact renters through decreased LTR rents. They can also positively affect new home buyers through decreased housing prices. However, restricting STRs can negatively affect homeowners through decreased housing prices. These results are stronger in areas with higher STR activity pre-restrictions (e.g., "touristy" areas).
- Spatial restrictions (e.g., prohibiting STRs in one community but not a bordering community) can cause a reduction STRs in communities with the spatial restrictions and an increase STRs in communities adjacent to the communities with the restrictions. The distribution effects are then different by community.
- In addition to affecting prices, restrictions on STRs can reduce housing investment, particularly for new additions (e.g., accessory dwelling units)
- Increasing the operating costs of STRs (e.g., through increased taxation or fees) can reduce STR activity, but there are distributional effects.
- Increasing the operating costs of STRs may reduce smaller-scale home sharing activity (by hosts who are more likely to be economically disadvantaged) and increase the relative proportion of commercial STR activity.

Overall, the empirical evidence suggests that STRs and STR regulations have an effect on the housing market; however, there is nuance. On the one hand, STRs can negatively impact renters through reduced LTR vacancies (particularly of smaller, more affordable units) and higher rental prices, and can negatively impact new home buyers in the short run through increased house prices. On the other hand, STRs can also positively impact homeowners through increased house prices and potentially positively impact renters in the long run through increased affordable units (e.g., ADUs). Policy makers considering STR regulations should be aware of these distributional effects and be prepared to face both the positive and negative consequences of them.

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1. Introduction

In this document, we examine the empirical literature related to STRs. We begin by looking at the empirical literature that measures the impact of STR activity on different aspects of the housing market. Following that, we examine the literature on how various policy tools impact STRs and different aspects of the housing market. The different aspects of the housing market we consider are: long-term rental (LTR) rents, LTR vacancy, housing prices, and spatial distributions. Policy tools we examine (because they have empirical evidence) are: restrictions on STRs, including personal residence restrictions and spatial restrictions, and STR policy tools that increase the cost of operating an STR (e.g., increased taxes, tourism levies, license fees, etc.). The majority of the empirical evidence examined is international: there are very few empirical Canadian studies of the impact of STRs and STR regulatory tools on the housing market. Regardless, the empirical evidence can inform us of potential impacts of STR regulations in Calgary.

There are several take-aways from a review of the empirical evidence:

1. An increase in STR activity have been shown to have a positive causal impact on LTR prices and housing prices. The magnitude of this effect differs, but it generally suggests that increases in STR activity explains about from five to 20 percent of LTR rent increases and 14 percent to 33 percent of housing price increases.
   - In areas with more STRs, units that shift to STRs are more likely to be “affordable housing units” (e.g., smaller units with less amenities) and units that would otherwise have sat vacant (e.g., whose hosts are away part-year so the unit could not be used as an LTR). Hosts of these units (e.g., otherwise vacant, small units) are more likely to be economically disadvantaged.
2. Restrictions on STRs can significantly reduce STR listings. However, there is no evidence whether/what proportion of those STR listings are put on the LTR market, sit vacant, or become owner-occupied.
3. Restrictions on STRs affect LTR and house prices differently: there are distributional effects.
   - Restricting STRs (e.g., outright prohibitions and/or personal residence requirements) can positively impact renters through decreased LTR rents. They can also positively affect new home buyers through decreased housing prices. However, restricting STRs can negatively affect homeowners through decreased housing prices. These results are stronger in areas with higher STR activity pre-restrictions (e.g., “touristy” areas).
   - Spatial restrictions (e.g., prohibiting STRs in one community but not a bordering community) can cause a reduction STRs in communities with the spatial restrictions and an increase STRs in communities adjacent to the communities with the restrictions. The distribution effects are then different by community.
4. In addition to affecting prices, restrictions on STRs can reduce housing investment, particularly for new additions (e.g., accessory dwelling units)
5. Increasing the operating costs of STRs (e.g., through increased taxation or fees) can reduce STR activity, but there are distributional effects.
   - Increasing the operating costs of STRs may reduce smaller-scale home sharing activity (by hosts who are more likely to be economically disadvantaged) and increase the relative proportion of commercial STR activity.

Overall, the empirical evidence suggests that STRs and STR regulations have an effect on the housing market; however, there is nuance. On the one hand, STRs can negatively impact renters through reduced LTR vacancies (particularly of smaller, more affordable units) and higher rental prices, and can negatively impact new home buyers in the short run through increased house prices. On the other hand, STRs can also positively impact homeowners through increased house prices and potentially positively impact renters in the long run through increased affordable units (e.g., ADUs). Restricting STRs then can result in both "winners" and "losers", with "winners" being persons currently in the rental market and current home buyers, and "losers" being persons who currently own their homes and renters who will be in the rental market in the future. Likewise, increasing STR operating costs through increased taxes and fees has the potential to decrease STR activity, but it can also increase commercial STR activity to the detriment of smaller, potentially more economically disadvantaged hosts. Policy makers considering STR regulations should be aware of these distributional effects and be prepared to face both the positive and negative consequences of them.

2. The Effect of STR Activity on the Housing Market

There are a handful of empirical studies that have tried to quantify the effect of STRs on the housing market. In this section, we summarize those papers, examine the high-level takeaways, and discuss what the results suggest for policy. Before delving into the papers, it should be noted that quantifying the causal effect of STRs on the housing market is difficult. In areas where STRs are increasing (both in number of listings and demand), these are the same areas more likely to see population growth, higher housing demand and rising rents, and gentrification. This results in a positive association between STRs and housing prices as they are moving in the same direction; however, this correlation may be spurious. Causal models aim to look beyond associations and find the causal mechanisms: does an increase in STR supply *cause* an increase in housing prices and/or LTR rents?

We group papers first by method used then by question looked at. The first group of papers uses variations in STRs across geography and time to examine the causal impact of STRs on the housing market. This strand of literature can tell us about the effect of STRs on the housing market in general. The second group of papers exploit variations in restrictive STR regulations to examine the causal impact of restricting STRs on the housing market. This strand of literature can tell us what we can maybe expect if STRs were restricted. For both groups of literature, we look at what the papers have to say about the impact of STRs on house prices and on LTR rents. In the second group of papers, we also examine the impact of STR restrictions on STR activity generally.

2.1 The Impact of STRs on the Housing Market

2.1.1 Long-Term Rents

In this section, we examine papers that examine how the growth of STRs have affected LTR rents. Table 1 summarizes the empirical evidence in this area.

The first paper by Barron, Kung, and Proserpio (2021) looks at the effect of STRs on LTR rents using a comprehensive data set across all of the US, exploiting geography and timing of STR activity. Using STR listings and rent at the zip code-year-month level and an instrumental variable strategy,[1] they find that in zip codes with the median owner-occupancy rate (72 percent), a 1 percent increase in STRs leads to a 0.018 percent increase in LTR rents. This decreases in the owner-occupancy rate: at zip codes at the 25th and 75th percentile owner-occupancy rate, a 1 percent increase in STRs increases rents by 0.024 percent and 0.014 percent respectively. This decrease in the owner-occupancy rate potentially occurs because in areas with less owner-occupiers, there is a higher likelihood of changing from long-term to short-term rentals. In the median zip codes, these estimates suggest that STRs account for an annual $9 monthly increase in rents, about one-fifth of actual rent growth.

In the second paper, Garcia-Lopez et al. (2020) examine the effect of STRs on LTR rents in Barcelona, Spain, exploiting geography and timing of STR activity. Using a number of empirical methods, they find that an increase of 100 STRs increases LTR rents by 3.5 percent. From 2012-2015, the average increase in STRs per neighborhood is 54 suggesting an increase of 1.89 percent in LTR rents. This effect is larger for tourist areas where STR activity is at the highest: STR effects in the most touristic parts of the city are substantial, increasing rents by as much as 7 percent. Garcia-Lopez et al. (2020) argue that although the effects are not small, STRs cannot explain the bulk of the large increase in rents from 2012 to 2016 (which increased on average by 38 percent).

In the third paper, Koster, van Ommeren, and Volkhausen (2021) examine the effect of STRs on LTR rents in Los Angeles county, US. Using an IV method,[2] they find that an increase of the share of Airbnb's to houses by one percent increases the LTR rents by 4.9 percent. Given that this finding is secondary to the main purpose of their paper, they do not explore it further.

The last paper in the table is the paper by David Wachsmuth (2023) who looks at the effects of STR on LTR rents for the province of British Columbia in Canada. This paper is different from the others in that it examines association, not causation. Regardless, he finds that an increase of one dedicated STR per 100 dwelling units increases LTR monthly rents by $0.49. Overall, he finds that from 2017-2019 STRs accounted for 19.8 percent of actual LTR rent increase. Average monthly rents in BC grew by $72 so STRS contributed on average $17 of that growth.

[1] The instrument used is worldwide google searches for Airbnb from 2008 – 2016 (before the take-off of Airbnb in most areas) combined with how "touristy" a zip code is, proxied by the number of food service and accommodation establishments in the zip code.

[2] The instrument used was whether or not a city has in place a home sharing ordinance (HSO) or not. Whether or not this is a credible instrument is debatable: it is possible that HSO's were put in place because of higher LTR rents.

Table 1: Paper Summary: Effect of STRs on LTR Rents

Paper	Geography	Effect of STRs on LTR Rents	How much of Rent Increase Explained by STRs
Barron, Kung, and Proserpio (2021)	US	In zip codes with the median owner-occupancy rate (72%), a 1% increase in STRs leads to a 0.018% increase in rents. Higher where owner-occupancy rate is lower.	Explains about one-fifth (20%) of actual rent growth
Garcia-Lopez et al. (2020)	Barcelona, Spain	An increase of 100 STR increases LTR rents by 3.5%. Higher in touristy areas.	Explains about 5% of rent increase (on average)
Koster, van Ommeren, and Volkhausen (2021)	Los Angeles County, US	A 1 percentage point increase in the share of Airbnb to homes increases rents per square metre by 4.9%. Higher in touristy areas.	Not stated.
Wachsmuth (2023) (note: not causal)	British Columbia, Canada	An increase of 1 dedicated STR per 100 dwelling units increases LTR monthly rents by $0.49.	STRs accounted for 19.8% of actual LTR rent increase

The take-away from this literature is that, overall, STRs have a positive, statistically significant impact on LTR rents in the locations of study (US, Barcelona, BC, LA). On average, STRs contribute from five percent to 20 percent of actual rent increases (with the remaining amount explained by factors other than STRs). These effects are larger in areas with higher STR activity (and smaller in areas with less STR activity).

2.1.2 Housing Prices

In this section, we examine papers that examine how the growth of STRs have affected house prices. Table 2 summarizes the empirical evidence in this area. The three papers examined in this section are the same as the papers examined in the previous section, thus we omit the methodological details in the discussion here.

In the first paper, Barron, Kung, and Proserpio (2021) In zip codes with the median owner-occupancy rate (72 percent), a 1 percent increase in STRs leads to a 0.026 percent increase in housing prices. This decreases in the owner-occupancy rate: at zip codes at the 25th and 75th percentile owner-occupancy rate, a 1 percent increase in STR increases housing prices by 0.037 percent and 0.019 percent respectively. This pattern is the same as was seen with LTR rents, but the effect is larger: STRs are estimated to have a larger impact on house prices than on LTR rents in the Barron paper. In the median zip codes, this accounts for an annual $1,800 increase in house prices and about one-seventh of actual housing price growth.

In the second paper, Garcia-Lopez et al. (2020) find that increase of 100 STRs in Barcelona, Spain increases housing (transacted) prices by 8.5 percent. From 2012-2015, the average increase in STRs per neighborhood was 54 suggesting an increase of 4.59 percent in housing (transacted) prices (the average increase in transacted prices was 15 percent). Compared to the effect of STRs on LTR rents (discussed in the previous section), the effect of STR on house prices is larger. Further, these effects were larger in areas with more STR activity: STR effects in the most touristic parts of the city are substantial, increasing housing (transacted) prices by as much as 17 percent (total transacted price increases in these areas was 28 percent).

In the final paper in this section, Koster, van Ommeren, and Volkhausen (2021) find that a 1 percentage point increase in the share of Airbnb to homes increases housing price (per square metre) by 3.0 percent. A standard deviation increase in the share of Airbnbs to homes is associated with a 5.5 percent increase in housing prices, which the authors argue suggests this implies effect of Airbnbs is substantial. Note, however, that this is the only paper of the three that finds a smaller impact of STR on house prices compared to LTR rents. Overall, they find that for LA county as a whole, Airnbnb's have increased average property values by 3.6 percent from 2008 – 2018. Effects are bigger in tourist areas: STR increases house prices by 14.7 percent if within 5km of Hollywood Walk of Fame and by 20 percent if within 2.5k of Walk of Fame. Overall, the nominal house prices within 5km of Walk of Fame have increased by 40 percent over the same period, suggesting Airbnb's explain about one-quarter of the nominal price increase.

Overall, the take-away from literature is that STRs have a positive, statistically significant impact on housing prices in the locations of study (US, Barcelona, LA). In two of the three papers examined, the researchers find that the STRs have a larger impact on housing prices than on LTR rents. On average, STRs contribute about 14 percent to 33 percent of actual housing price increases. These effects are larger in areas with higher STR activity.

Table 2: Paper Summary: Effect of STRs on House Prices

Paper	**Geography**	**Effect of STRs on House Prices**	**How much of House Price Increase Explained by STRs**
Barron, Kung, and Proserpio (2021)	US	In zip codes with the median owner-occupancy rate (72%), a 1% increase in STRs leads to a 0.026% increase in housing prices. Higher where owner-occupancy rate is lower.	Explains about one-seventh (14%) of actual house price growth
Garcia-Lopez et al. (2020)	Barcelona, Spain	Increase of 100 STRs increases housing (transacted) prices by 8.5%. Higher in touristy areas.	Explains about 30% of house price increase (on average)
Koster, van Ommeren, and Volkhausen (2021)	Los Angeles County, US	A 1 percentage point increase in the share of Airbnb to homes increases housing price (per square metre) by 3.0%	Airbnb's explain about ¼ of the nominal house price increase for houses within 5km of Walk of Fame

2.1.1 Evidence on Who Shifts to STRs

As suggested, in the previous section, we saw that there is evidence that an increase in STRs increases LTR rents. The most likely explanation is that STRs remove LTR from the housing market, increasing LTR rents. However, in some cases, otherwise vacant spaces are also converted to STRs, making more efficient use of space and serving as an additional source of income. Li, Kim, and Srinivasan (2024) examine the question of what types of units are converted to STRs and the demographics of the different hosts using a structural empirical model of STRs and LTRs in the US. They confirm the evidence suggested by the previous section, finding that LTRs are shifting to STRs (the “cannibalization effect”). They also find that properties that hosts would not have listed as LTRs (e.g., because the host lives in the dwelling part-time) also shift to the STR market—what the authors call the “market expansion effect”. They find that the market expansion effect is larger than the cannibalization effect in cities with many STRs, such as New York and Miami, and for “affordable units”, e.g., smaller STRs with fewer bedrooms and amenities. Furthermore, they find that otherwise empty units that are converted to STRs have hosts that are more likely to be economically disadvantaged, e.g., lower income. This research suggests that while LTRs are being converted to STRs (and many of these are also affordable units), this effect is smaller than the number of units that would otherwise sit empty being converted to STRs. The combination of these effects harms local renters by reducing affordable LTRs, but it also provides an income source for local hosts with units that would be otherwise vacant and who are economically disadvantaged.

Based on these findings, Li, Kim, and Srinivasan (2024) suggests a “convex tax” on STRs, e.g., an accommodation tax that set at a higher rate for more expensive STRs and a lower rate for STRs in affordable housing units/less expensive STRs. This results in fewer LTRs being converted to STRs and it has a smaller impact on economically disadvantaged hosts (while falling more heavily on economically advantaged hosts).

2.2 Examining the Effect of STR Restrictions

2.2.1 LTR Rents and Housing Prices

In this section, we summarize the empirical evidence that examines the effect of STR restrictions on LTR rents and housing prices. While the papers in the previous section examined the effect of the proliferation of STRs on the housing market, this section examines what happens in the housing market when STRs are restricted. Focusing on empirical causal evidence, we look at two papers.

In the first paper by Koster, van Ommeren, and Volkhausen (2021), they examine the effect of home sharing ordinaces (HSO’s). HSOs out-rightly prohibited STRs in some cities, while limiting STRs to personal residences in other cities (with the remainder of cities not passing any form of HSOs). The researchers use a spatial regression discontinuity design to examine the effect of HSO on LTR rents and house prices. First, they find that HSOs caused strong reduction in Airbnb listings: Airbnb listings in cities with an HSO declined by 6.1 percentage points or about 20 percent of listings. This differed by STR type: HSO’s caused a reduction in entire home/apartment Airbnb listings by 50 percent, with a smaller effect on STR room listings (and no effect on room listings where personal residence STRs were still permitted). Second, they found that HSOs reduce housing prices (per square metre) by 1.8 percent and reduce LTR rents (per square metre) by 2.3 percent (with stronger effects in areas with more STR activity), indicating STRs are effective at reducing housing market prices. However, the authors also argue that HSOs also have distributional effects. The “winners” of HSOs are LTR renters who see LTR rents decreases while the “losers” are homeowners who see their housing values decrease.

In the second paper, Valentin (2020) examines the effect of a prohibition on STRs in the French Quarter of New Orleans in 2016. This prohibition was actively enforced by Airbnb by an agreement between the City and the platform. Using a differences-in-discontinuity method, this paper finds that the STR prohibition in the French Quarter caused a reduction in STR listings in the French Quarter: STR listings in the French Quarter were 138 percent more likely to be removed as listing compared to STR listings in other communities in New Orleans. Further, the prohibition in the French Quarter caused spillovers into adjacent neighborhoods: adjacent neighborhoods saw an increase in the number of STR listings, higher STR demand, higher prices per unit, and higher revenue per unit. Finally, the prohibition in the French Quarter was found to have caused a reduction in housing prices by 30 percent, a loss of about $77 per square foot. This paper did not examine the effect of the prohibition on LTR rents.

Table 3: Paper Summary: Effect of STR Restrictions on LTR Rents House Prices

Paper	**Geography**	**STR Restriction**	**Effect of Restriction on STR Listings**	**Effect of STR Restrictions on LTR Rents**	**Effect of STR Restrictions on House Prices**
Koster, van Ommeren, and Volkhausen (2021)	Los Angeles County US	Home Sharing Ordinance	HSO caused 6.1 percentage points or about 20% reduction of listings. Reduced entire home listings by 50%	HSOs reduce LTR rents (per square metre) by 2.3%	HSOs reduce housing prices (per square metre) by 1.8%
Valentin (2020)	New Orleans, Louisiana, US	Prohibition of STRs in French Quarter	138% more likely to be removed as listing in French quarter	N/A	Prohibition caused reduction in housing prices by 30%,

Overall, the main take-away from this body of research suggests that, at least in the areas studied, STR restrictions can impact the housing market. STR restrictions can decrease STR listings, LTR rents and house prices. This is consistent with the empirical evidence examined in section 1.1 that suggested that an increase in STRs can increase LTR rents and house prices. However, it also suggests there are distributional impacts that must be taken into consideration.

2.2.2 Housing Investment

There is also empirical evidence that examines the effect of STR restrictions on housing investment (measured as the change in building permits in a given locality). Bekkerman et al. (2023) examines this issue for the US with a focus on Los Angeles country. Exploiting differences in timing of STR restrictions (the HSO's previously discussed), they find that restricting STRs causes a reduction in STR listings. More importantly, STR restrictions cause a reduction in permits for new constructions by 18 percent, and new additions (particularly ADU's) by 17 percent. Overall, the authors argue that STRs (or conversely, the restriction of STRs) increase investment in the housing market. In particular, the reduction in ADU housing permits suggests that STR restrictions may reduce the construction of affordable rental units in the long run.

3. The Effect of Increased STR Fees on STRs and the Housing Market

In a set of papers, Garz and Schneider (2023a, 2023b) examine the effect of taxes on STRs. In both Denmark and Norway, STR operators are required to report STR revenues to the country's tax authority for tax purposes. In Denmark, the government entered into a data sharing agreement with Airbnb requiring Airbnb to report revenues made on Airbnb listings along with social security numbers to the Danish tax authority to improve tax compliance. Norway did not have a data sharing agreement with Airbnb, instead relying on Airbnb operators to self-report their revenues.

Using a causal method, Garz and Schneider (2023a) show that in Denmark when Airbnb actively participated in enforcement of taxes, this reduced the number of STR listings listed by single-listing hosts, increased the price of STRs listed by single-listing hosts, and reduced STR bookings for single listing hosts. Comparatively, multi-listing hosts increased their prices, increased their availability, and increased the number of booked days. In addition, there was a spatial re-organization from urban to rural STRs: hosts in low-STR-activity areas (e.g., rural) expanded the number of listing days whereas hosts in high STR-activity areas (e.g., urban) tended to make their property less often available. Overall, increased tax compliance coupled with active enforcement caused higher commercialization of the STR market (e.g., an increase in the proportion of multi-listings hosts as single listing hosts exited). This same effect was not seen in Norway where higher taxes were *not* actively enforced: the taxes had no effect on the number of composition of STRs (Garz and Schneider 2023b).

Taxes on STR revenues increase the operating cost of STRs however payment of these taxes are difficult to track by tax authorities. These papers provide causal evidence that suggests that increases in operating costs of STRs may only be effective if coupled with active enforcement.

4. Bringing it Together

Overall, the empirical evidence supports in general the perception that the increased growth of STRs has a positive casual effect on LTR rents and housing prices. While the increased growth of STRs cannot explain the bulk of the LTR rental and housing prices increases seen, they can explain up to about one-third of the price increases. While restricting or prohibiting STRs is perceived as the "obvious" policy response, there are important nuances that must be considered. Restricting STR's will have distributional effects: there will be "winners" and "losers". While restricting STRs has the potential to decrease LTR rents thereby benefitting current renters, it will also decrease housing prices, having a potentially negative impact on current homeowners who would like to sell their homes (although this may be beneficial for new home buyers who face lower house prices). Further, restricting STRs may harm economically disadvantaged hosts whose units would otherwise remain vacant but for the option of STRs. Finally, restricting STRs may reduce housing investment, particularly investment in secondary suites and ADUs, potentially harming renters in the future. Policy makers should be aware of these trade-offs and be prepared to face the consequences of their policy decisions.

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